Real-world use of a personal wearable exoskeleton in individuals with spinal cord injury

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Introduction

Spinal cord injury (SCI) often results in loss of standing and walking ability, affecting mobility, health, and social participation. Wearable lower-limb exoskeletons have shown safety, feasibility, and usability in supervised rehabilitation settings [1], but evidence on long-term home/community use remains extremely limited, with only one previous real-world study in individuals with SCI [2]. This study aims to characterize the real-world use, user experience, and user-device interaction of the ABLE Daily, a personal lower-limb exoskeleton specifically designed for home and community environments.

Materials and Methods

This pilot study is prospective, observational, and descriptive, and is designed to evaluate the ABLE Daily: a wearable robotic lower-limb exoskeleton for personal use by individuals with SCI, specifically intended for home and community environments, that actively assists hip and knee flexion–extension through motorized joints (Figure 1). Up to five adults with chronic SCI will be included according to predefined criteria, including lesion level T1–L5, AIS A–D, WISCI 0–9, and previous experience with the robotic exoskeleton. Each participant must have a trained companion to assist during home/community use.

The study follows four stages: (1) a 2–3 day baseline phase without the device, during which usual activity, physiological context, pain, spasticity, perceived health, and daily-life events are recorded using ActiGraph LEAP™, questionnaires, VAS scales, and logbooks; (2) nine 90-minute training sessions in a clinical setting; (3) a community-skills qualification test; and (4) a 3-month home/community phase with a personalized ABLE Daily device.

Training includes basic and advanced skills required for daily-life use, such as donning/doffing, sit-to-stand and stand-to-sit transitions, indoor/outdoor walking, turning, ramp negotiation, slalom walking, door handling, elevator use, and approaching a chair to sit. The qualification test verifies that participant and companion can safely perform key community tasks with at least minimal assistance before home use.

The primary outcome is user experience, including usability, acceptability, comfort, perceived safety, and subjective impact. Secondary variables include training requirements, usage patterns, activity and physiological context, recurrent incidents, perceived changes, and user/companion feedback. During home/community use, participants will be encouraged to use the device three times per week for 30 minutes when possible, while documenting context, purpose, obstacles, incidents, perceived exertion, pain changes, and device performance. Biweekly follow-ups will support logbook completion and early detection of usability or safety issues. Data analysis will be descriptive and exploratory, combining quantitative summaries with qualitative analysis of recurrent difficulties and feedback.

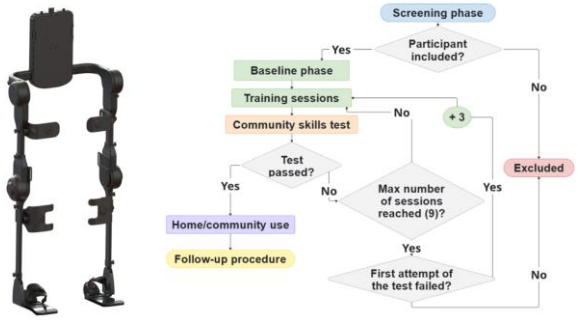


Figure 1: ABLE Daily lower-limb exoskeleton (left) and schematic overview of the study workflow (right).

Expected Results

This study aims to identify barriers and facilitators for personal use, describe how users and companions acquire operational autonomy, and determine where, why, and how often the device is used in domestic and community settings. Preliminary results will be presented at the conference. Based on previous literature [2], we hypothesize that use will mainly concentrate on exercise, upright mobility, and social participation, while limitations related to transportability, support needs, and everyday functional tasks may remain relevant. We also expect participants to progressively acquire operational autonomy, with relatively fast learning curves and increased daily-life engagement. The study is expected to provide practical evidence on learning, ergonomics, and human-device interaction, informing future device design, training protocols, and larger-scale studies.

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References

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2. van Dijsseldonk R. et al., Sci. Rep., 2020